

TECHNICAL DOSSIER

SwarmX

Fleet of cooperative autonomous drones based on distributed control, autonomous navigation, and swarm algorithms.

DOSSIER DATA

PROJECT LEAD

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STATUS

Technical planning

REPOSITORY

github.com/RoboTech-URJC/Swarm-X

LAST UPDATE

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Project Description

SwarmX is a project focused on developing a fleet of autonomous drones capable of working cooperatively through distributed control technologies and swarm algorithms. The proposal aims to build an experimental aerial platform in which several unmanned vehicles can communicate with each other, coordinate trajectories, and execute joint missions in controlled environments.

The project combines mechanical design, electronic integration, embedded-systems programming, artificial intelligence, device-to-device communications, and validation through simulation and flight testing. Its educational goal is for participants to gain a complete view of autonomous aerial vehicle development, from structure fabrication to the implementation of navigation and coordination strategies.



Objectives

General objective: design, build, and program a fleet of autonomous drones capable of collaborating with each other through distributed communication and swarm algorithms.

Specific objectives:

- Design and 3D-print structural components for multirotor drones.
- Integrate Pixhawk flight controllers, brushless motors, GPS, cameras, batteries, and auxiliary electronics.
- Implement autonomous navigation, obstacle detection, and communication between nodes.
- Use tools such as ROS2 and advanced processing platforms to coordinate multiple drones.
- Validate fleet behavior through simulation and testing in controlled environments.
- Explore social-impact applications, such as emergency support or distribution in remote areas.

Development Program

Phase 1: 3D design and printing. The first phase will focus on modeling and printing the drone's structural components, validating geometries, mounts, and internal component distribution.

Although the initial proposal considers PLA for prototypes, the material planning includes PETG, ASA Aero, and PC FR to improve the resistance of final parts and increase safety in critical areas.

Phase 2: electronic integration. The Pixhawk 6C flight controller, brushless motors, GPS, batteries, ESCs, camera, and auxiliary processing systems will be integrated.

This phase will make it possible to validate power distribution, propulsion assembly, sensor readings, and basic communication between the drone's different electronic modules.

Phase 3: programming and autonomy. The drones will be programmed with tools such as ROS2 and advanced processing platforms to implement autonomous navigation, obstacle detection, and communication between nodes.

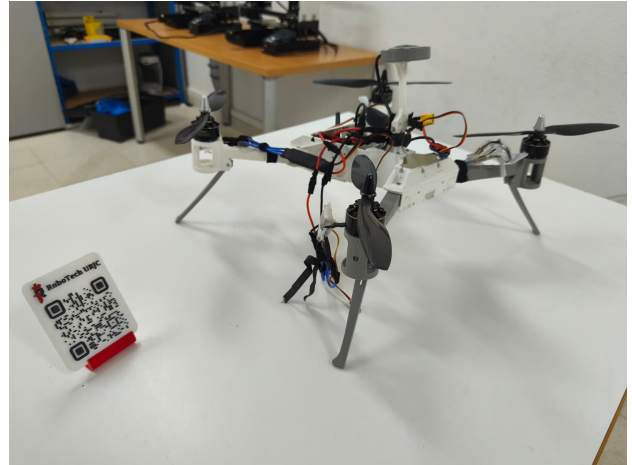
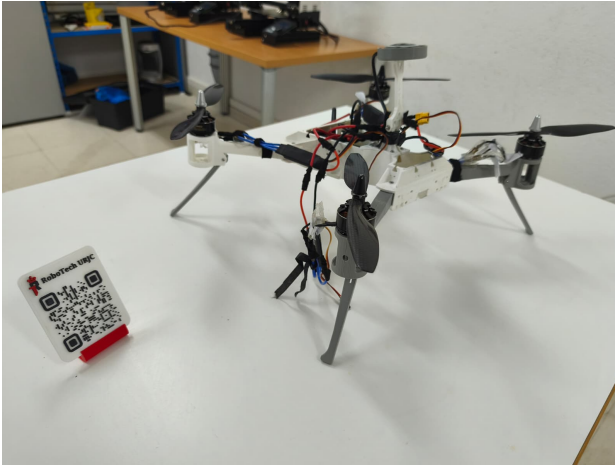
The software architecture must allow each drone to act as a node within the fleet, sharing relevant information and responding to coordination commands.

Phase 4: simulation and flight testing. Tests will be carried out in controlled environments to evaluate the individual and collective behavior of the fleet.

During this phase, stability, navigation, communication, and swarm-coordination parameters will be adjusted to move toward safe cooperative flights.

System Architecture

SwarmX is planned as a distributed system. Each drone will have its own flight controller, power system, positioning sensors, and communication capability. On top of this base, an additional processing layer will execute navigation, perception, and coordination algorithms.



Communication between drones will be a fundamental part of the project. The goal is not only to make one drone fly autonomously, but to create a foundation that enables several vehicles to coordinate and study collective behaviors similar to those of a swarm.

Applications and Project Value

SwarmX has high educational value for the association because it brings together several advanced areas of robotics: aerial vehicle design, digital fabrication, power electronics, flight control, computer vision, artificial intelligence, and distributed communications.

In addition, the project has potential social applications. A fleet of cooperative drones could eventually be used to support emergency operations, inspect hard-to-access areas, assist in search tasks, or help distribute small resources in remote areas.

Next Goals

- Define the initial mechanical architecture of the drone.
- Review compatibility between Pixhawk 6C, motors, batteries, ESCs, and sensors.
- Design the first printable structural parts.
- Prepare the simulation and development environment with ROS2.
- Assemble a first functional prototype for individual tests.
- Plan communication between nodes for future swarm coordination.